

Simple Calculator – Project Documentation

1. Introduction

This project is a **Simple Calculator Web Application** developed using **HTML, CSS, and JavaScript**. It performs basic arithmetic operations such as addition, subtraction, multiplication, and division. The calculator is designed with a clean user interface and simple logic, making it suitable for beginners and mini-project submissions.

2. Objectives

- To understand the structure of an HTML document
- To design a basic user interface using CSS
- To implement calculator logic using JavaScript functions
- To perform arithmetic operations dynamically on user input

3. Technologies Used

- **HTML** – To create the structure of the calculator
- **CSS** – To style and align the calculator layout
- **JavaScript** – To handle user interactions and calculations

4. Project Structure

The calculator consists of the following main components:

- Input display area
- Numeric buttons (0–9)
- Operator buttons (+, −, ×, ÷)

- Clear (C) button
- Equal (=) button

5. Code Explanation

5.1 HTML Section

- The `<input>` field displays the numbers and results.
- `<button>` elements are used for digits and operators.
- The `onclick` attribute calls JavaScript functions when buttons are pressed.

5.2 CSS Section

- `body` uses flexbox to center the calculator on the screen.
- `.calculator` class styles the calculator box with padding, border-radius, and shadow.
- Buttons and input fields are styled for better readability and usability.

5.3 JavaScript Section

appendValue(value)

- Appends the clicked number or operator to the display.

clearResult()

- Clears the display when the C button is clicked.

calculate()

- Evaluates the expression entered by the user using `eval()` and displays the result.

6. Working Principle

1. User clicks number and operator buttons.
2. The input field displays the entered expression.
3. On clicking the = button, the expression is evaluated.
4. The calculated result is displayed instantly.
5. The C button clears the display for a new calculation.

7. Sample Output

- Input: $8 + 2$
- Output: 10
- Input: $9 * 5$
- Output: 45

8. Advantages

- Simple and user-friendly interface
- Fast calculation
- Easy to understand and modify

9. Limitations

- Uses `eval()` which is not recommended for large or secure applications

- Supports only basic arithmetic operations

10. Conclusion

This Simple Calculator project demonstrates the basic integration of HTML, CSS, and JavaScript. It is an excellent beginner-level project to understand DOM manipulation and event handling in JavaScript.